SOP 5

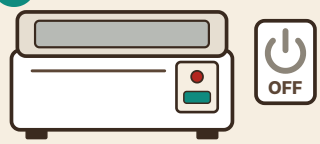
Weekly maintenance and calibration

When: Every week (Friday after the last job), after a tube change, and after moving the laser

Time: About 45 min (Estimate) **Who:** Owner

You need: Lens tissue and lens cleaner · Brush and vacuum · Distilled water · Fit coupon file · A scrap sheet · The log at the end of this SOP

1

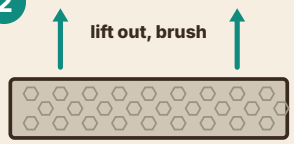


Switch off and unplug.

Let the tube and head cool for 10 minutes.

- Exhaust off, vent closed.

2



Clean the honeycomb and tray.

Lift out the honeycomb, brush off resin, empty and vacuum the tray under it.

- No scraps left under the bed.

3

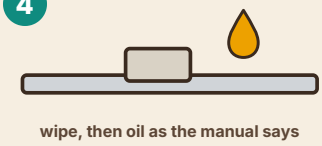


Clean mirrors 1, 2, 3 and the lens.

Lens cleaner on lens tissue, one wipe per tissue. Put each part back the same way round.

- Cracked, pitted or brown: replace it.

4

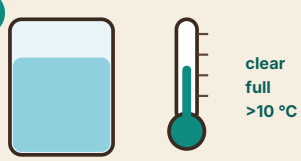


Wipe the rails and belts.

Dry cloth on the rails. Oil only where and when the OMTech manual says.

- The head slides smoothly by hand.

5

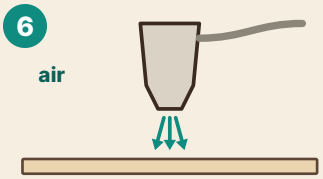


Check the water loop.

Level, clear water, no algae. Top up with distilled water. Change it as the manual says.

- Above 10 °C. In winter keep the heat on.

6

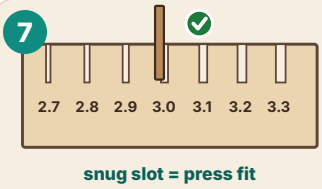


Check air assist and exhaust.

Feel the air at the nozzle. Look in the duct for soot and grease. Clean the fan monthly.

- Steady air. Duct clear.

7

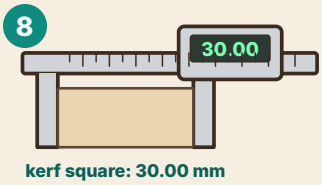


Cut the fit coupon on this week's wood.

File `fit_test_coupon` in `kits/coupon`. Push a 3 mm strip into the slot comb. Note the snug slot.

- Snug slot moved 0.1 mm or more: update the kerf.

8

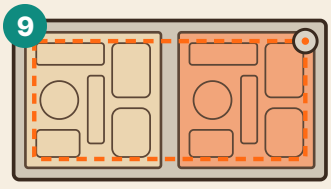


Measure the kerf squares.

The square part should be 30.00 mm, the hole 20.00 mm. Also try the dowel holes and the 0.3 to 0.6 mm bridges.

- Changed: set `KERF` in `config.py`, then run `rebuild_all.sh`.

9

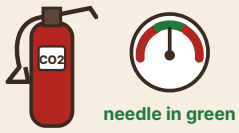


Check the camera and frame.

Frame a shape on a scrap. The red dot should land where the camera image shows.

- Off by more than 1 mm: recalibrate the camera in LightBurn.

10



Check the fire kit.

Extinguisher needle in green, pin and seal in place, blanket on its hook.

- The smoke alarm beeps when tested.

Weekly log

Tick each box when done. Write the numbers from the coupon. Keep this page at the laser.

WEEK OF	BED AND TRAY	MIRRORS AND LENS	RAILS	WATER	AIR AND EXHAUST	SNUG SLOT (MM)	KERF (MM)	FIRE KIT	INITIALS

Hartwood Designs, SOP-5, Rev A, 2026-10-12. Times and settings marked Estimate are planning numbers until measured on the real machine. Machine details follow the OMTech manual when it differs from this sheet.